

Owner's Project Requirements

Technical & requirements

Program	ATLAS Habitat Initiative
Asset	ATLAS 01 (first habitat)
Status	Draft v0.1 — for review by Architect/Record + Professional Engineers
Scope	Canada (launch: Ontario assumed) !' United States !' international
Companion docs	Legal & Compliance · Requirements (PRD) · Business Plan

****Disclaimer.**** This OPR is an internal design intent document, ****not a permit set and not engineering certification.**** Every system must be designed, stamped, and approved by licensed professionals and the Authority Having Jurisdiction (AHJ) before construction. ATLAS is ****operationally independent, not legally exempt**** — all codes apply in full.

1. Owner's intent

A self-sufficient residential habitat: one floor per human need (water, energy, food, shelter, air, health, restoration), with a basement reclamation core and a rooftop amenity. The building generates power, recycles water/waste, and grows part of its food. **Life safety never depends on AI or cloud** — see §4.

2. Functional program (per level intent)

Level	Use	Key owner requirements
Basement	Reclamation core: reservoir, water treatment, anaerobic digester/ biogas	Code-rated containment, gas detection, isolation; potable/ non-potable separation
Energy	ESS (battery), inverters, controls	ESS in a fire-rated room (not an open wall): rated walls, vented louvers, deluge
Water	Treatment, dual plumbing	CSA B128 dual plumbing; purple non-potable riser + backflow preventer
Food	Vertical farm / aquaponics	Food safety handling; isolated micro-livestock if any (never open farming)
Shelter	Dwelling floors	Unit/bed counts per program; barrier-free units; fire separation
Air	Atrium "Lung" / ventilation	IAQ, pressurization integration
Health	Commons + telehealth + accessible WC	Keep out of Group B (no clinical bays) to avoid hospital occupancy

Restoration	Skydeck / pool	Roof pool only; parapet + fall protection
Rooftop	PV array, reservoir, comms mast	Structural + electrical + lightning/grounding

3. Code compliance matrix (verify per AHJ)

Domain	Governing code/standard (CA launch)	Owner requirement
Structure	National Building Code of Canada + provincial (e.g. OBC); IBC based in US	PE stamped; 3D printed shell as accelerator with engineered alternative solution
Fire & life safety	OBC Part 3; NFPA where referenced	2 pressurized exit stairs + vestibules; firefighter elevator (OBC 3.2.6); areas of refuge; sprinkler/standpipe risers
Egress / accessibility	OBC barrier free; Accessible Canada Act / AODA; ADA (US)	Barrier free units + paths; areas of refuge designed in
Elevators	Provincial elevator code (e.g. TSSA)	Passenger cars A/B + protected firefighter car
Electrical / ESS	CE Code; battery storage standards	Licensed design; ESS fire rated room; utility interconnection even if grid is backup
Potable water	Provincial drinking water regs	Multi stage treatment; monitored, logged water quality
Greywater/blackwater reuse	Plumbing code; provincial reuse approvals	Reuse class standards; non potable clearly separated
Biogas / digester	Environmental permits; fire code; emissions	Engineered containment, detection, pressure relief
Food production	CFIA / Safe Food for Canadians; provincial public health	Inspection + traceability if sold
Zoning	Municipal	Pre application meetings; frame food/energy as building services

4. The safety boundary (non negotiable)

Deterministic, safety rated controls (BMS/PLC) hold fire, egress, electrical isolation, and biogas safety. **ATLAS OS / AI / cloud is observe and advise only** and must never be a prerequisite for any life safety function. This boundary is a design requirement and a regulatory approval asset.

5. Permits & approvals checklist (initiate early)

- Zoning pre application / variances
- Building permit (PE/architect stamped set; alternative solution package for 3D print)
- Electrical permit + utility interconnection/net metering agreement
- Plumbing permit (dual plumbing, backflow)
- Drinking water system approval + sampling plan
- Wastewater/greywater reuse approval
- Environmental permit (digester/biogas) + fire code review

- Food premises / CFIA registration (if selling produce)
- Elevator licensing (incl. firefighter car)
- Fire department plan review + occupancy permit

6. Commissioning & verification

Independent commissioning of water quality, ESS, fire/egress, and the safety boundary.
Telemetry logs (water quality, energy, incidents) retained for audit and SR&ED.

Pair this OPR with stamped Basis of Design documents from each discipline before P1 close.

